

Yousef A. Hassan

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OBJECTIVE

Computer Science graduate with full-stack project experience across TypeScript, Python, PostgreSQL, Linux, API testing, data analysis, and distributed systems.

EDUCATION

Worcester State University

B.S. Computer Science; Minor in Physics; GPA: 3.06

Worcester, MA

May 2026

CLASS PROJECT EXPERIENCE

CircuitNorth Commerce

2026

Next.js / TypeScript / PostgreSQL / Supabase

- Built a responsive technology storefront with catalog search and filters, product variants, persistent cart, demo checkout, customer orders, cancellation, and account workflows.
- Created a role-based operations console for product, category, variant, inventory, discount, customer, fulfillment, refund, CSV report, and audit workflows.
- Designed versioned Zod-validated APIs, Supabase RLS, and transactional stock functions; passed 9 Vitest and 10 Playwright desktop/mobile tests plus strict TypeScript, ESLint, accessibility, and production-build checks.

Retail API Contract Tests

2026

Python / pytest / HTTPX / JSON Schema

- Automated CircuitNorth API contracts for successful requests, validation errors, role denial, conflicts, missing resources, required headers, and response schemas.
- Passed 7 isolated MockTransport tests at 90% coverage and 3 live integration tests against the running Next.js API.

Retail Analytics Dashboard

2026

Python / pandas / Streamlit / Plotly

- Built an interactive dashboard that removes duplicate and invalid rows, calculates revenue and order metrics, and analyzes category, customer, daily-sales, and inventory trends.
- Verified cleaning and aggregation logic with 6 pytest tests and 98% package coverage; confirmed the Streamlit service through its runtime health endpoint.

Linux Systems Toolkit

2026

Bash / WSL / Bats / ShellCheck

- Created command-line utilities for filtered log summaries, permission audits, process thresholds, disk reports, and file-integrity manifests with machine-readable output.
- Passed ShellCheck and 7 Bats tests in Ubuntu, including invalid-input, changed-file, and threshold-validation cases.

Retail Database & Reporting Lab

2026

PostgreSQL / SQL / pgTAP

- Designed a normalized retail schema with generated money totals, historical order snapshots, constraints, reporting views, targeted indexes, deterministic seed data, and query-plan examples.
- Implemented atomic inventory reservation that rejects overselling and records movements; passed all data-quality gates and 22 pgTAP assertions on PostgreSQL 14.

Cloud Job Queue

2026

FastAPI / Celery / Redis / Docker

- Built an idempotent report API and responsive queue console with queued, running, completed, failed, and dead-letter states, metrics, retries, and recovery controls.
- Verified separate API, Redis, and Celery worker processes end to end; passed 8 API tests at 84% coverage, including transient retry and three-attempt dead-letter behavior.

FarmData2 Fork Contribution

2025

JavaScript / Docker / Git

- Contributed an ONBOARDING.md command correction in a course fork through commit 5de07e3 on the fix-typo-drush branch without claiming ownership of the upstream application.
- Practiced repository navigation, isolated branch work, documentation review, and traceable Git history inside a larger JavaScript and Docker project.

Relevant Technical Coursework

2022-2026

Computer Science Coursework

- Data Structures; Algorithm Analysis; Database Design & Applications; UNIX Systems Programming; Operating Systems; Computer Networking & Security; Data Mining; Big Data Analytics Capstone; Cloud/Parallel/Distributed Computing; Software Construction/Design/Architecture; Computer Organization & Architecture; Probability & Statistics; Technical Writing.

SKILLS

Languages: TypeScript, Python, C/C++, JavaScript, SQL, Bash/Shell

Web & Data: Next.js, React, FastAPI, REST, OpenAPI, PostgreSQL, Supabase, pandas, Streamlit, Plotly

Systems & Testing: Linux/Unix, Git/GitHub, Docker, Redis, Celery, GitHub Actions, pytest, Vitest, Playwright, Bats, ShellCheck

AWARDS

Dean's List

Fall 2022, Spring 2026